



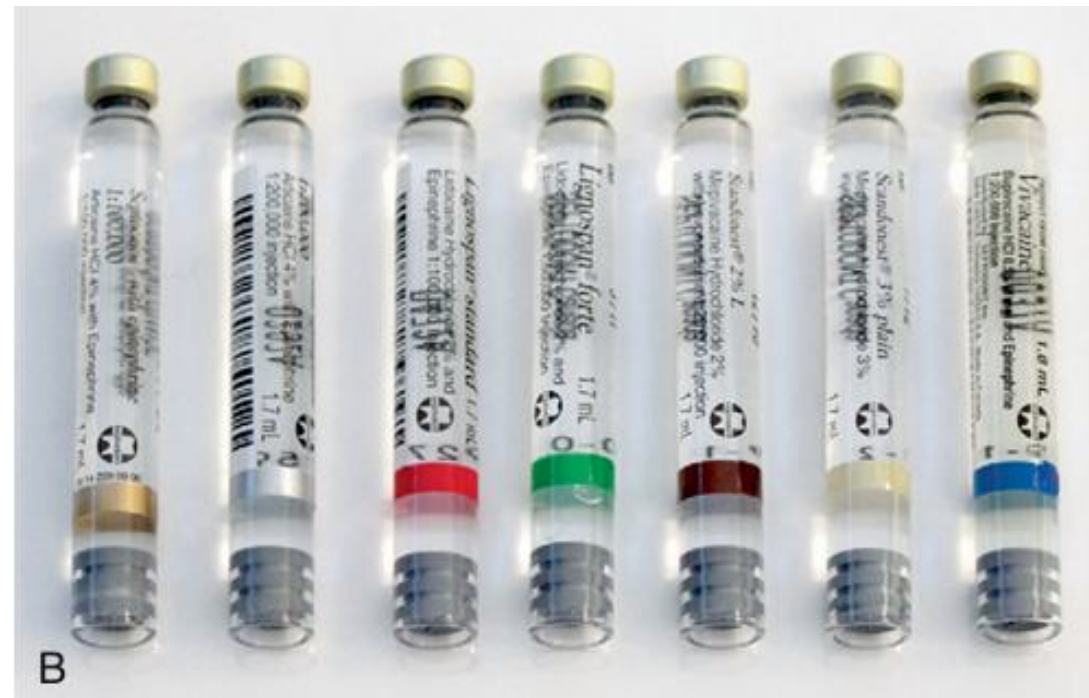
Oral Surgery

Lecture 6

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“Pharmacology of Local Anesthesia” part1



Constituents of local anesthetic solution (Local anesthetic cartridge):

- 1. local anesthetic agent** (esters or amides type)
- 2. Vasoconstrictor agent.** (will be discussed later)

3. Reducing agent: vasoconstrictors in local anesthetic solution are unstable and **may oxidize**, especially on prolonged exposure to sunlight. This will lead to **brown discoloration** of the solution and this is an indication that the solution should be discarded.

In an attempt to overcome this problem a small quantity of **sodium bisulphite**, which compete for the available oxygen, is included in the solution. Since this substance is more readily oxidized than adrenaline or nor adrenaline, **it protects their stability.**

4. **Preservative:** the sterility of local anesthetic solution is maintained by the inclusion of a small amount of preservative. Some preservatives such as **methyl paraben** have been shown to produce **allergic reaction** in sensitized subjects.

5. **Fungicide:** in the past some solutions tended to become cloudy due to the proliferation of minute fungi, now a small quantity of **thymole** is added to serve as a fungicide and prevent this occurrence.

6. **Vehicle:** the anesthetic agent and the additives mentioned above are dissolved in **modified ringer's solution**. This isotonic vehicle minimizes discomfort during injection.

7. **Distilled Water** it is used as diluting substance to provide the volume of solution in the dental cartridge.

1.Local anesthetic agents”

Local anesthetics are drugs which upon topical application or local injection **cause a reversible** loss of sensory perception, especially of pain in a restricted area of the body; it provide pain control during dental therapy and **interrupt the propagation of impulse preventing it from reaching the brain.**

- These drugs are listed by their percentage (%) concentration. The number of (mg) of an agent contained in the cartridge can be calculated by multiplying the percentage (%) concentration by the number of (ml) in the cartridge. Thus, a cartridge containing 1.8 ml of 2% local anesthetic solution contains 36 mg of the local anesthetic agent (2% mean that for each ml there is 20 mg, if the cartilage contains 1.8 ml then $1.8 \times 2 = 36$ mg of the local anesthetic agent in a single cartilage).

Classification of local anesthetics

The local anesthetic agents can be classified in various ways

1 – on the basis of occurrence in nature

- Natural (e.g. cocaine)
- Synthetic (e.g. lignocaine)

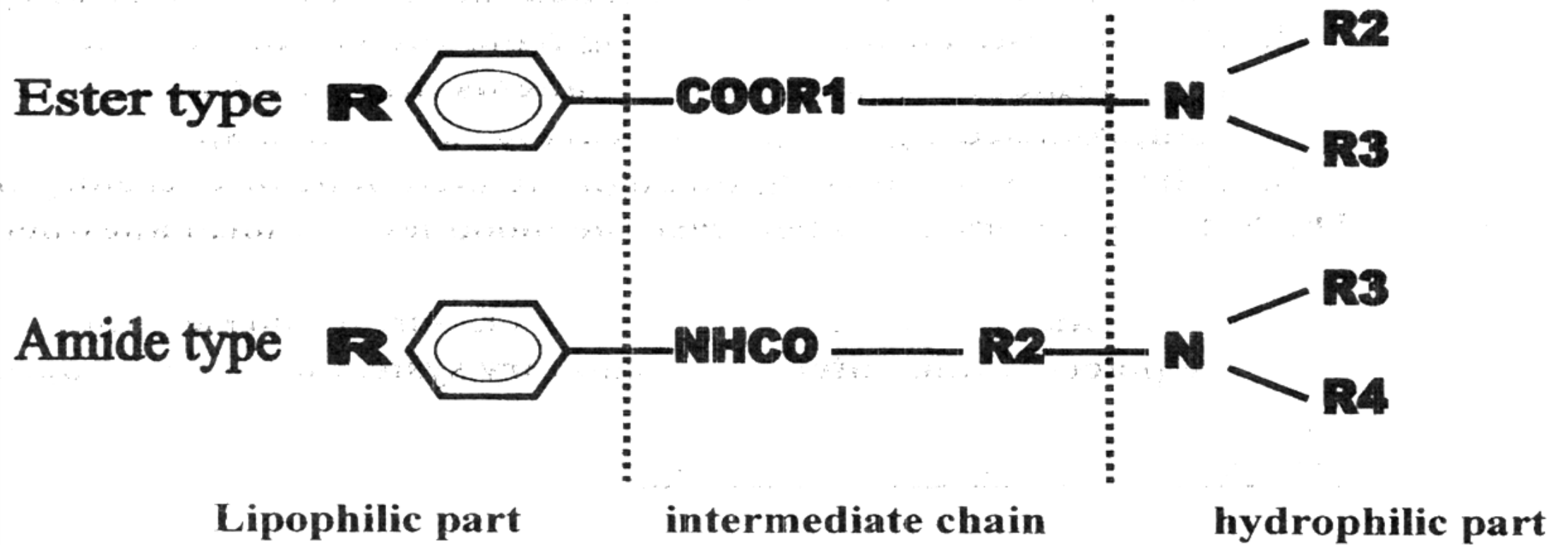
2 – on the basis of duration of action

- Short-acting (e.g. Articaine, lidocaine)
- Long-acting (e.g. bupivacaine)

3 – on the basis of chemical structure

- Ester (e.g. Procaine, Cocaine, Benzocaine)
- Amide (e.g. Lidocaine, Prilocaine, Articaine)

Local anesthetics are classified as either esters or amides according to their chemical linkages the chemical structure is shown below:



The **lipophilic** part is the largest portion of the molecule, which is aromatic in structure, while the **hydrophilic** part is an amino derivative of ethyl alcohol or acetic acid. The chemical structure is completed by an intermediate hydrocarbon chain containing **either an ester or an amide linkage**. All local anesthetics are amphipathic that is they possess both lipophilic and hydrophilic characteristics generally at opposite ends of the molecule.

Notes:-

1. local anesthetics without a hydrophilic part are not suited for injection but are good topical anesthetics (ex: benzocaine)
2. Ester linked local anesthetic (ex: procaine) are readily hydrolyzed in aqueous solution, while amide linked types (ex: lidocaine) are relatively resistant to hydrolysis.
3. A greater percentage of an amide linked drug is **excreted unchanged in the urine** than of an ester linked drug.

Pharmacokinetic of local anesthetics:-

1-Distribution:-

Once absorbed into the blood, local anesthetics are distributed throughout the body to all tissues. The blood level of local anesthetic is influenced by the following factors:-

1. Rate at which the drug is absorbed into the cardiovascular system.
2. Rate of distribution of the drug from the vascular compartment to the tissue (**more rapid in healthy patients than in those who are medically compromised e.g.; congestive heart failure patients**).
3. Elimination of the drug through the metabolic or excretory pathways.

The last two factors act to decrease the blood level of local anesthetic.

Note:-

All local anesthetics readily cross the **blood brain barrier**, they also readily cross the **placenta** and enter the circulatory system of the developing fetus.

2-Metabolism (biotransformation):-

A. Ester local anesthetics:-

Ester local anesthetics are hydrolyzed in the **plasma** by the enzyme **pseudo cholinesterase**. Procaine undergoes hydrolysis to **para amino benzoic acid** which is excreted unchanged in the urine. Allergic reactions that occur in response to ester drugs are usually not related to the parent compound (e.g.; procaine) but rather to **para amino benzoic acid** which is a major metabolic product of ester local anesthetics.

Peoples having atypical form of pseudo cholinesterase get inability to hydrolyze ester local anesthetics and other chemically related drugs thus get prolongation of higher blood levels of the local anesthetic and an increased potential for toxicity.

B. amide local anesthetics:-

The metabolism of the amide local anesthetics is more complex than that of the esters; the primary site of biotransformation of amide drugs is the **liver**. The metabolic products of certain local anesthetics are capable of producing significant clinical activity if permitted to accumulate in the blood (due to renal or cardiac failure and during periods of prolonged drug administration).

3-Excretion:

The **kidneys** are the primary organ of excretion for both the local anesthetics and its metabolites. A percentage of a given dose of local anesthetic drug will be excreted unchanged in the urine and this varies according to the drug. **Esters appear in only very small concentration as a parent in the urine;** this is because they are hydrolyzed almost completely in the plasma.

Amides are usually present in the urine as the parent compound in a greater percentage than are esters, because of their more complex process of metabolism of anesthetic compound or its metabolites from the blood, resulting in slightly elevated blood levels and an increased potential for toxicity. This may occur with either the esters or the amides drugs. Thus patients with a significant renal disease may represent a relative contraindication to the administration of local anesthetics.

- **Common local anesthetic agent**

Lignocaine (Lidocaine, Xylocaine)

Lignocaine: is the most commonly used local anesthetic agent in dentistry, it is the first non-ester type of local anesthetic agent to be used in dentistry, it is stable as it can be stored for a long time at room temperature, it withstands boiling and autoclaving and it is compatible with all types of vasoconstrictors.

- Classification: Amide.
- Metabolism: in the liver.
- Excretion: Via the kidneys. Less than 10% unchanged.

- The onset of Action: Rapid (3 to 5 minutes).
- Effective Dental Concentration: 2%.
- Anesthetic Half-Life: (90 minutes).
- Topical Anesthetic Action: Yes (with a concentration of 5% or 10%).
- Availability in dentistry: dental cartridge of 2% lidocaine with or without epinephrine

- **Maximum Recommended Dose:**
- The maximum recommended dose of **lidocaine** with epinephrine (1:100.000) is **(7 mg/kg)**. While **(4.5 mg/kg)** is the maximum dose of lidocaine without epinephrine. In all the cases, not to exceed an absolute maximum **dose of (500 mg)**.

- □ Allergy to amide local anesthetics is extremely rare; although possible, this is a major clinical advantage of lidocaine and all amides over ester-type local anesthetics.

❖ Selection of local anesthetic agent is based on:

1. Duration of action
2. Need for control of post-operative pain
3. Physical and mental status of the patient
4. Concomitant medications

❖ An approximate duration for completing the surgical procedure should be taken into account. Based on duration, the action of the local anesthetic agent can be grouped as:

1. **Ultrashort-acting agents:** where the duration of action is less than 30 minutes: e.g. 2 % lidocaine without vasoconstrictor.
2. **Short-acting agents:** where the duration of action is 30 to 90 minutes: e.g. 2% lidocaine with vasoconstrictor.

3. Medium-acting agents: where the duration of action is 90 to 180 minutes: e.g. 4 % prilocaine with vasoconstrictor.

4. Long-acting agents: where the duration of action is 180 minutes or longer: e.g. 0.5% of bupivacaine.

- ❖ Most of the oral surgical procedures result in varying amount of post-operative pain. The local anesthetic agent serves as additional medication that sometimes eliminates the need for postoperative analgesia.
- ❖ Any co-existing medical condition such as hypertension, diabetes should be considered, a patient with a history of allergy to a specific local anesthetic agent should be considered. The mental status of the patient has to be evaluated, small children and the mentally retarded patients may traumatize their lip, tongue or cheeks due to the effect of local anesthesia.

	The ester type	The amide type
1	Procaine	Lidocaine
2	Chloroprocaine	Prilocaine
3	Propoxycaine	Articaine **
4	Butacaine	Bupivacaine
5	Cocaine	Dibucaine
6	Benzocaine	Etidocaine
7	Hexylcaine	Mepivacaine
8	Piperocaine	
9	Tetracaine	

- ****Note:** articaine contains both ester and amide linkages in its chemical structure, but is considered as an amide local anesthetic agent.

A medical professional, wearing a white lab coat and a white surgical mask, is holding a large, clear medical syringe with a yellow plunger. The syringe is held in the person's right hand, with the thumb positioned on the plunger. The background is a blurred blue and white, suggesting a clinical setting. The text "Thank you" is overlaid on the image in a black, sans-serif font, enclosed within a thin blue rounded rectangular border.

Thank you